1 Bridge-Control — visual workflows

How each part of the rig works, from button to camera. Colors match the console areas (legend above the index).

REC / STOP
group and per-device recording

TIME
timecode, clocks, jam

CAMERAS
BMPCC, Canon, Sony, mobiles, GoPro

GATEWAYS
C3 (BMPCC) and S3 (GoPro)

ZOOM F6
FRC-8 link, card, jam

LOGS
shoot log, takes, SFTP

NETWORK
camera AP, uplink, web

SYSTEM
services, update, access

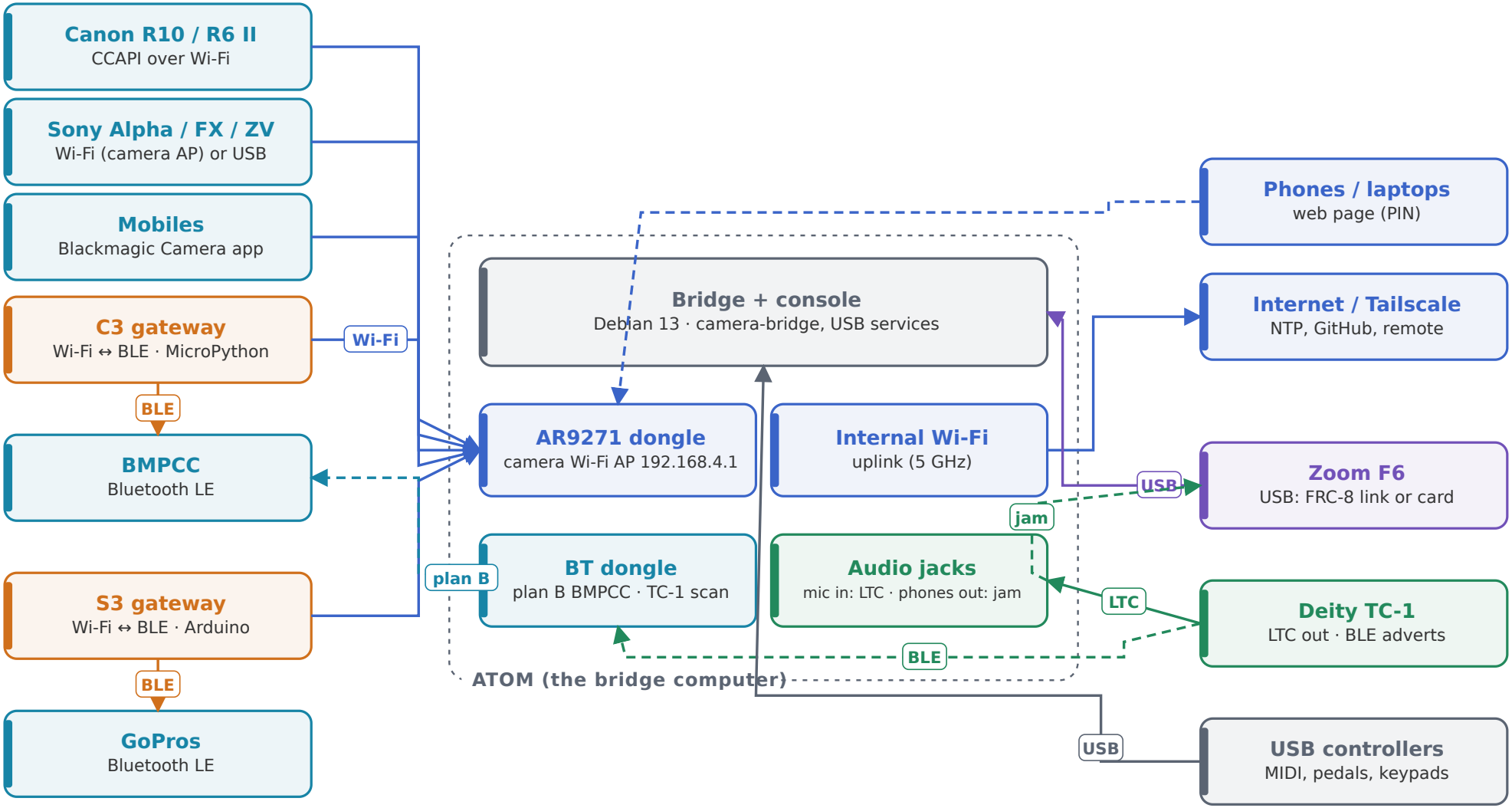
Pages

2 System map (hardware)
3 Software: services and ports
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- Read top to bottom, left to right. Solid arrows: commands and data. Dashed arrows: optional or fallback paths.
- Names are the ones used in the console and on the web page.

2 System map

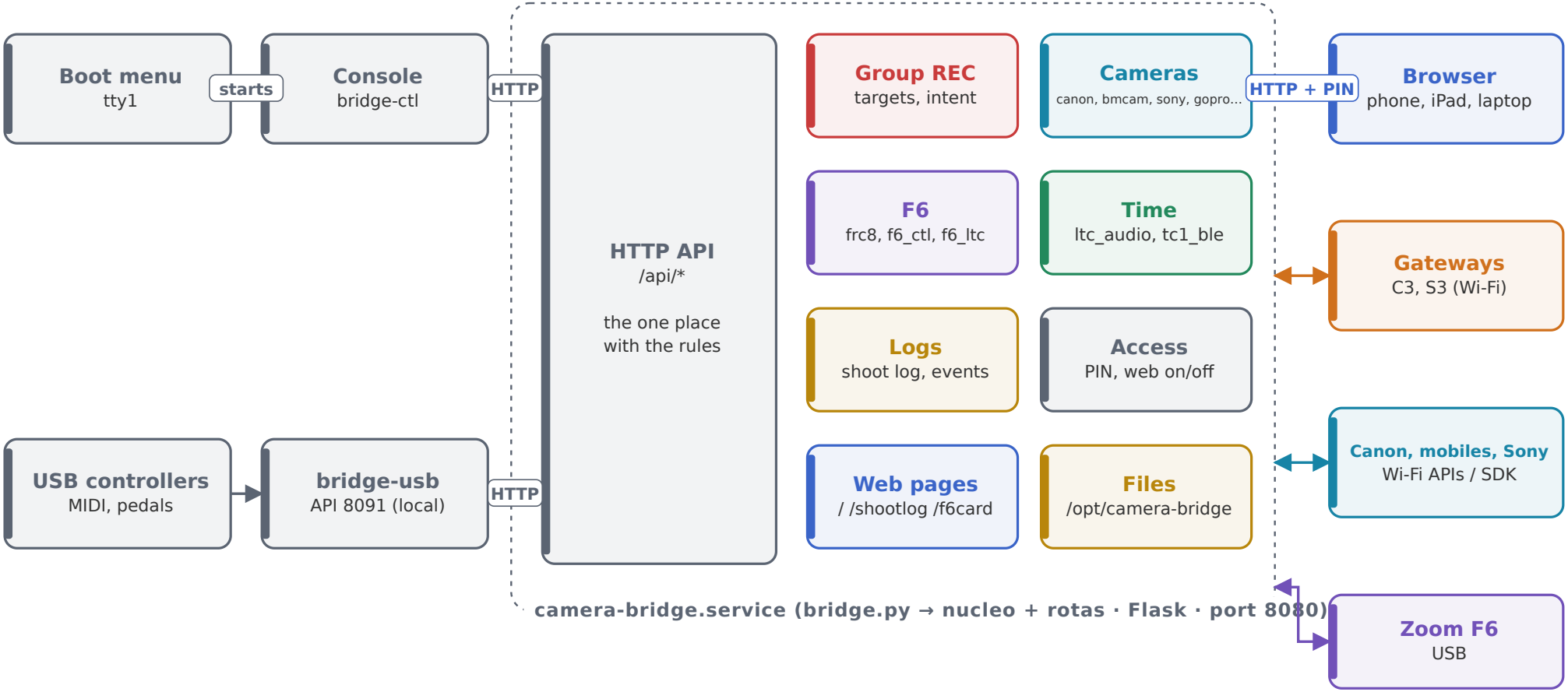
What talks to what. Everything on set meets on the camera Wi-Fi of the Atom's AR9271 dongle.



- Cameras without Wi-Fi control (BMPCC, GoPro) are reached through an ESP32 gateway that turns Wi-Fi commands into Bluetooth LE.
- The internal Wi-Fi is only the uplink (internet, NTP, GitHub, Tailscale); the camera network never depends on it.
- Plan B: the main BMPCC can be driven straight from the Atom's Bluetooth dongle when the C3 is not available.

3 Software: services and ports

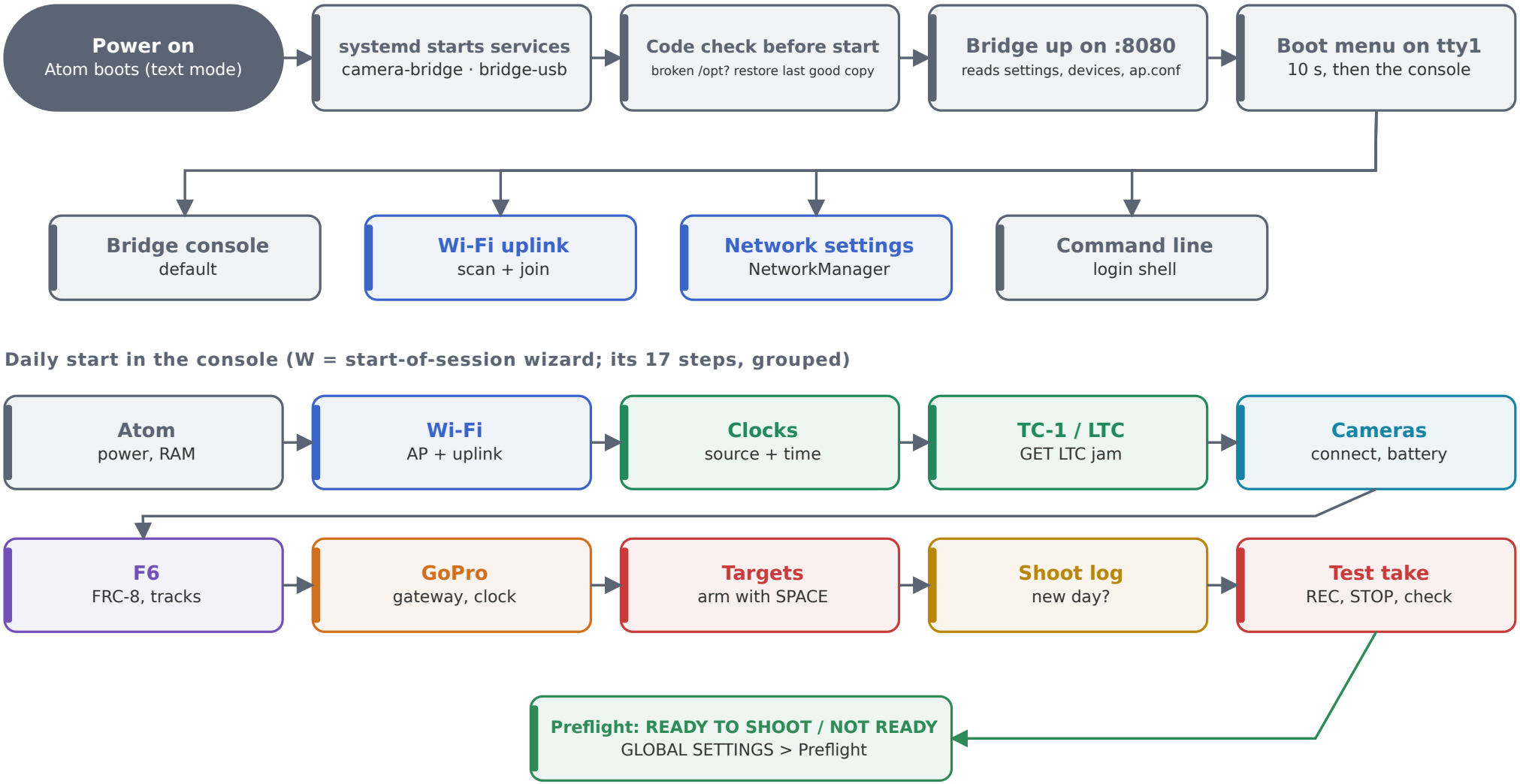
Two systemd services plus the bridge. Only the bridge talks to cameras; the console and USB use its API.



- The console, the USB service and the web page all go through the same HTTP API: one set of rules, no duplicated logic.
- The USB service and the web page for other devices can be switched off in GLOBAL SETTINGS > Services; the console keeps working.
- The local API (8091) only listens on 127.0.0.1; the console reaches it for its USB menus.

4 Power on and daily start

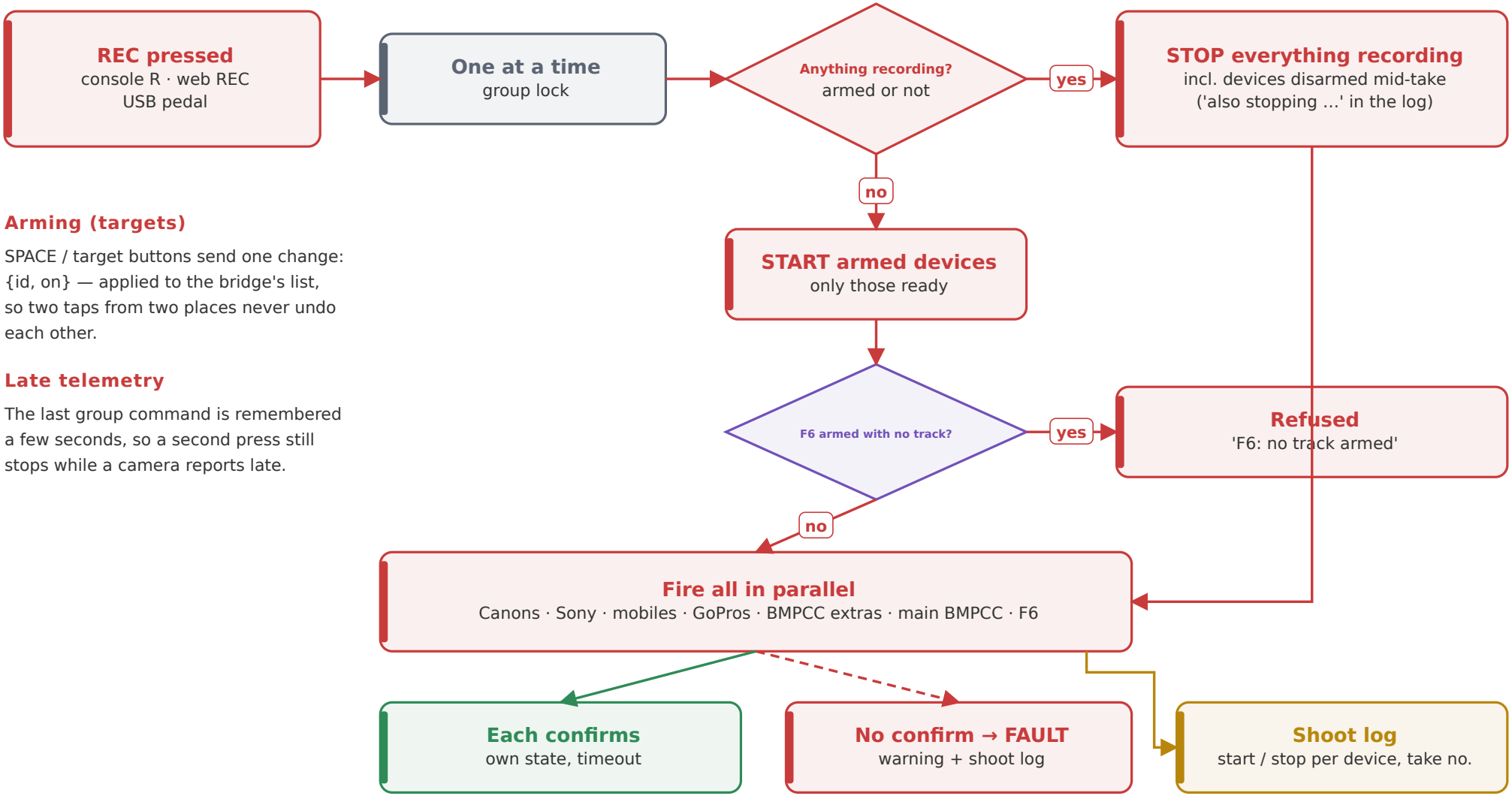
From the power button to the first test take.



- If the code in /opt is damaged (power cut during an update), the service restores the newest copy that compiles before starting.
- The console reopens by itself after an update, even with a page open.
- Preflight lists every check (cameras, time source, disk, memory, power) and gives one verdict, the same as the web page.

5 Arming and the group REC

One rule: if anything is rolling, the next press stops everything. Stopping is always one press away.



Arming (targets)

SPACE / target buttons send one change: {id, on} — applied to the bridge's list, so two taps from two places never undo each other.

Late telemetry

The last group command is remembered a few seconds, so a second press still stops while a camera reports late.

- Only devices in video mode and ready are started; skipped ones are named in the event log.
- A device that accepts the command but does not change state raises a visible FAULT and a 'fault' line in the shoot log.
- Per-device REC buttons (next page) never touch other devices.

6 REC on one device

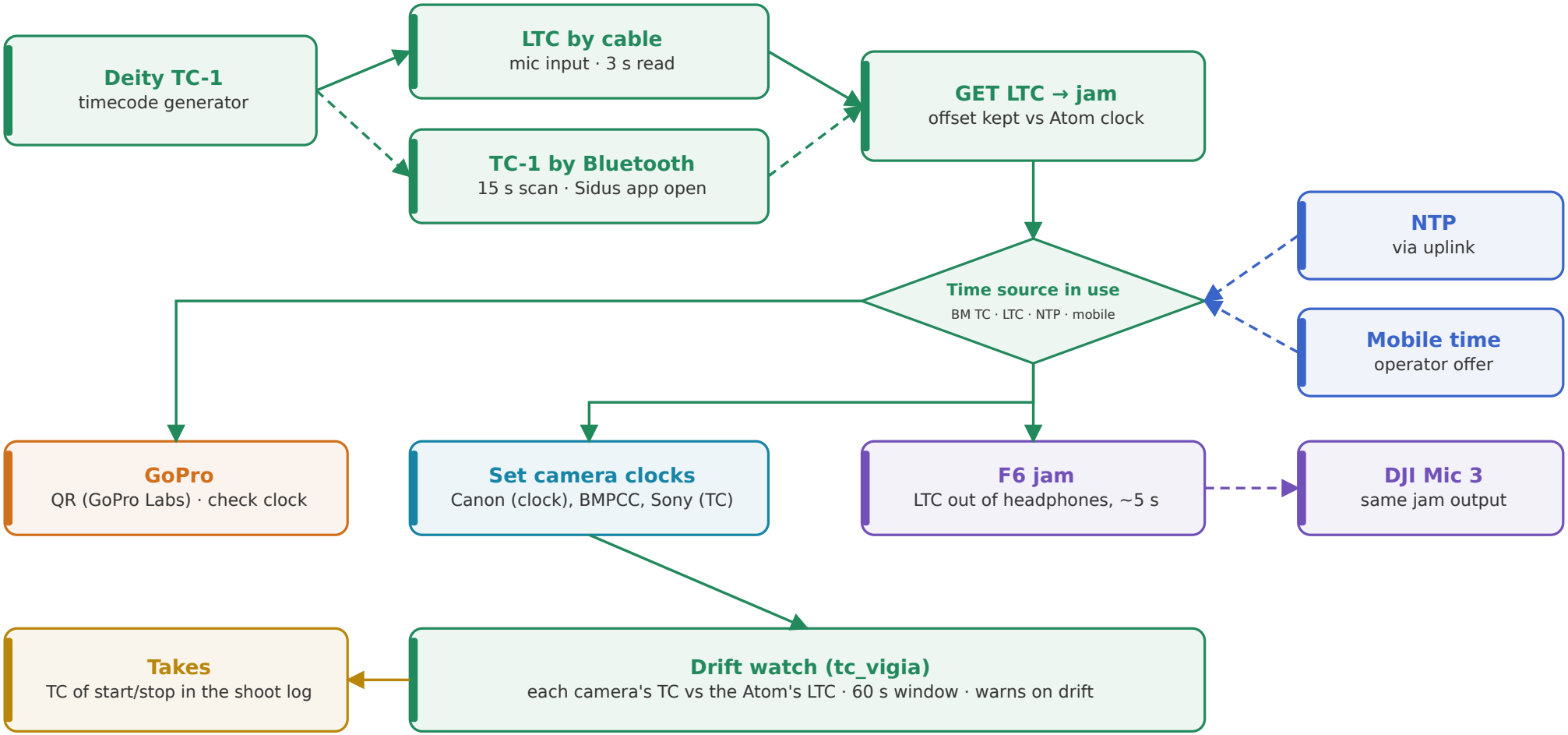
Console: T (or 2 in the option bar) on the device row · web: the device's REC button.



- A per-device REC only changes that device, and decides start or stop from that device's own state.
- On the GATEWAYS lines of the console, SPACE / T explain where the cameras are instead of doing nothing.
- While a device records, its REC option reads 'REC (recording: press to STOP)'.

7 Timecode and clocks

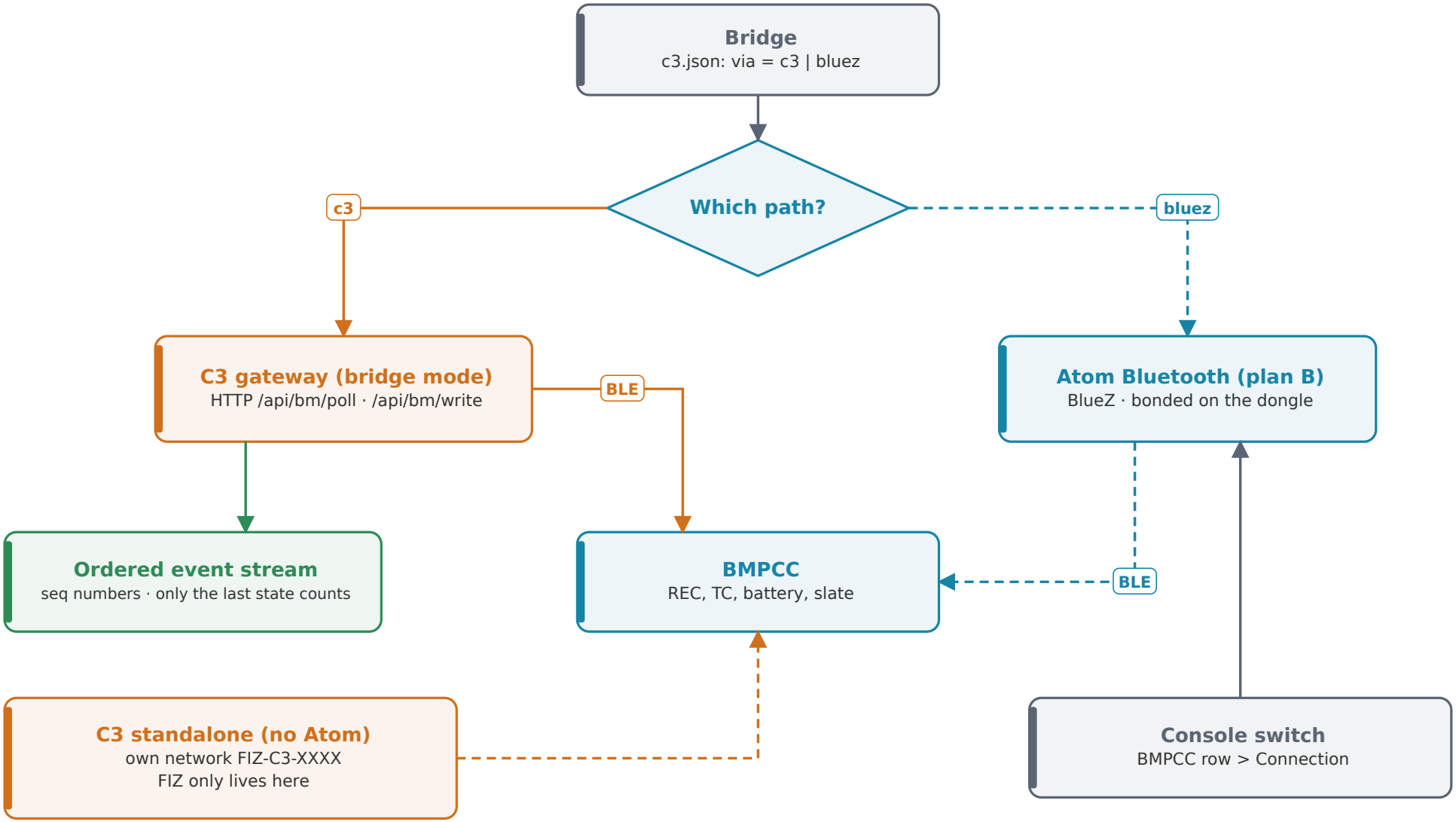
One reference (the TC-1), one jam on the Atom, everything else set or checked against it.



- The jam stores the difference between the TC-1 and the Atom clock; the Atom then free-runs (it has a real-time clock).
- Bluetooth reading (no cable) averages ~90 adverts over 15 s and corrects a measured +0.045 s offset.
- Nothing sets camera clocks automatically: it is always an operator action.

8 BMPCC: C3 gateway or Atom Bluetooth

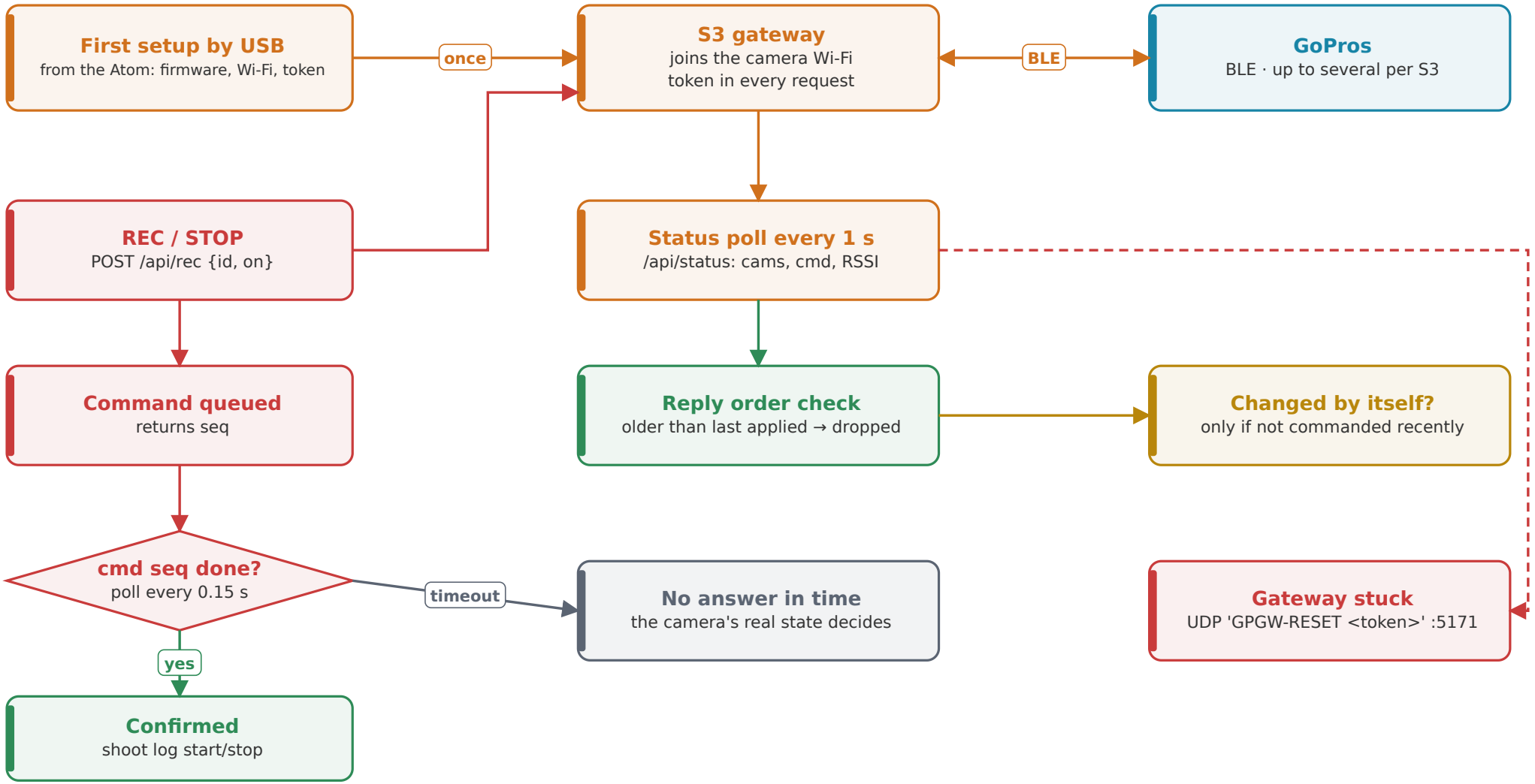
The main BMPCC goes through the C3 by default; plan B drives it from the Atom's own dongle.



- Switching path is refused while recording; the old session closes and the new one starts.
- The C3 reports memory and Bluetooth signal; low memory raises a warning (the firmware is precompiled to fit).
- Extra BMPCCs each have their own C3 (bm2, bm3 …) and use the same bridge-mode API.

9 GoPro through the S3 gateway

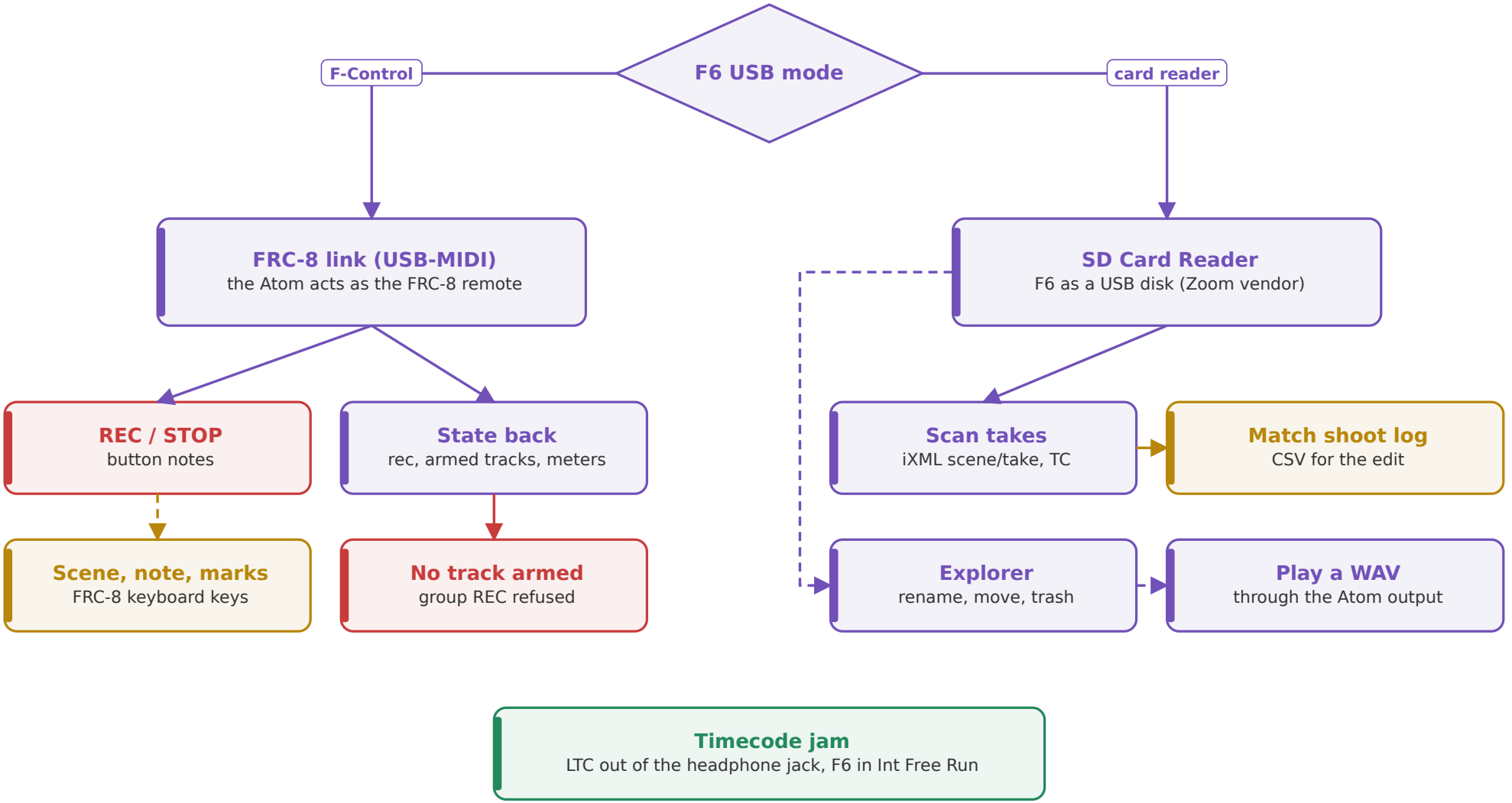
The S3 turns HTTP on the camera Wi-Fi into Bluetooth commands for the GoPros.



- The token is set only by USB / serial; every request on the Wi-Fi then carries it in the X-GW-Token header.
- Firmware updates always go through the Atom (GLOBAL SETTINGS > Update gateways).
- GoPro time: show the camera the QR from /gopro_qr (GoPro Labs); Check clock reads it back to about 1 s.

10 Zoom F6

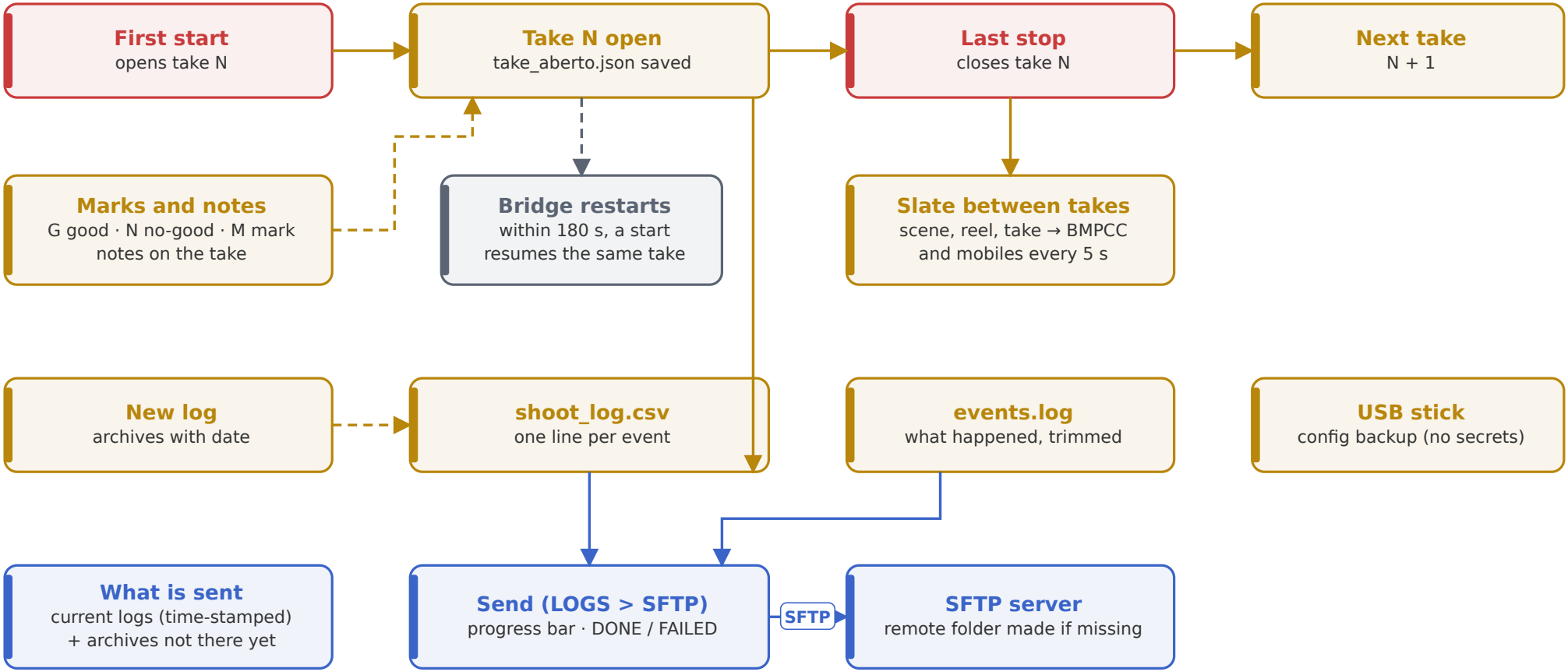
Two USB modes, chosen on the F6. Timecode comes in by audio, not USB.



- FRC-8 order that works: press connect on the F6 with the cable out, then plug it in.
- The card is only mounted during each operation; delete moves to F6_TRASH; a backup stick is never mistaken for the card (USB vendor check).
- If the F6 does not confirm REC/STOP within 1.5 s, a FAULT is raised.

11 Shoot log, takes, slate, send logs

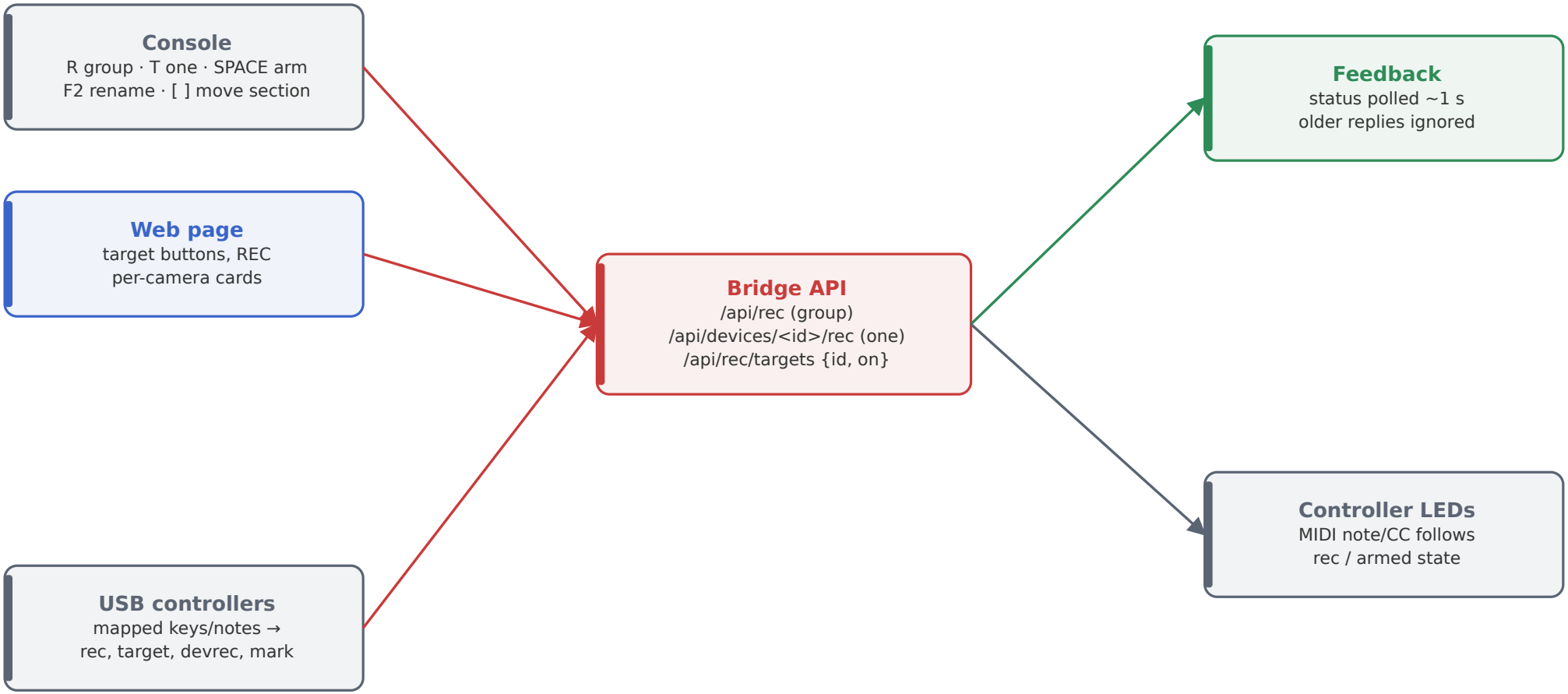
Every start and stop, from every device, lands in one CSV with the take number and TC.



- Take numbers restart at 1 only when you start a new shoot log (the old one is archived, never deleted).
- A REC/STOP pressed by the operator after a restart always opens a new take instead of resuming.
- The SFTP password is never shown or exported; it stays in a 0600 file on the Atom.

12 Control surfaces

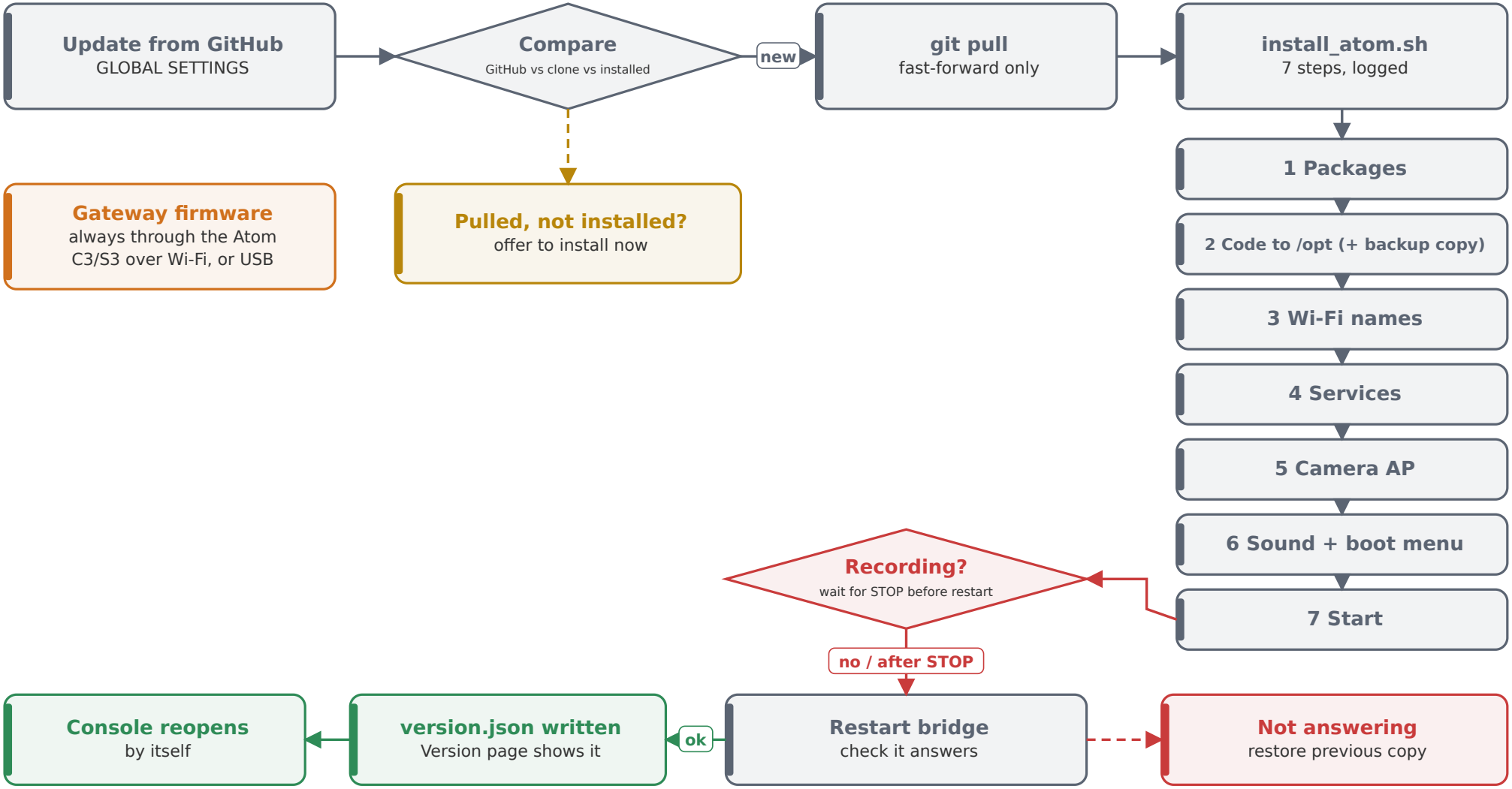
Three ways to drive the rig, one set of rules in the bridge.



- Fixed in this round: a status read that left before a tap and arrived after it no longer puts the old state back (page, console).
- Console menus: PgUp/PgDn move an item, [and] move a whole section, F2 renames an item or a section (kept across updates).
- Mappings for USB controllers live in usb_map.json and follow the device to any port.

13 Update from GitHub and gateway firmware

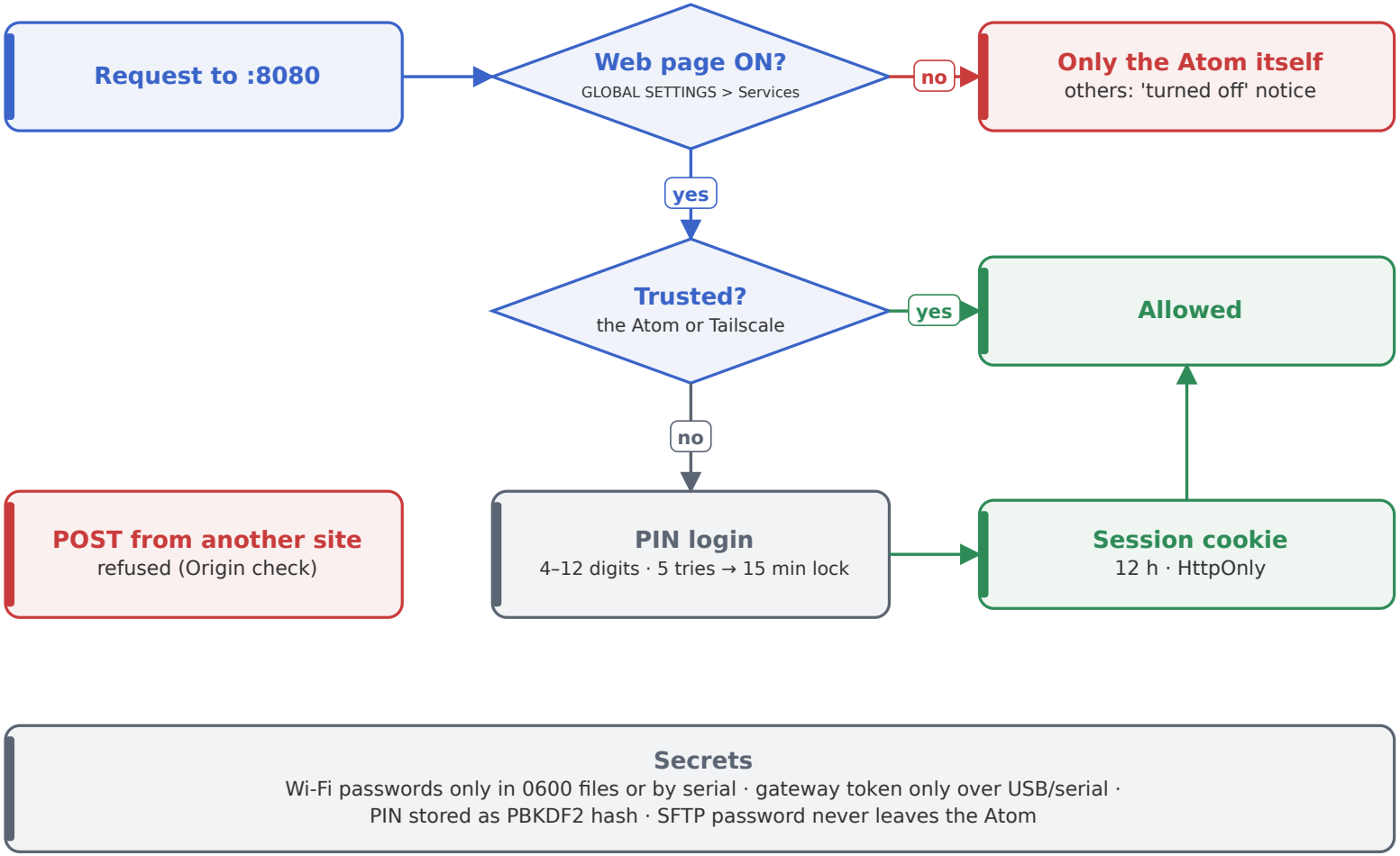
The Atom updates itself and then the gateways; nothing is installed half-way.



- The bridge is never restarted in the middle of a take: the installer waits for STOP (NO_WAIT_REC=1 skips it).
- The USB service stays off after an update if you switched it off.
- Gateway firmware: GitHub builds it, the Atom keeps it and sends it to each C3/S3 (Wi-Fi OTA, or USB for a blank board).

14 Access, services and warnings

Who can drive the rig, what can be switched off, and what the bridge warns about.



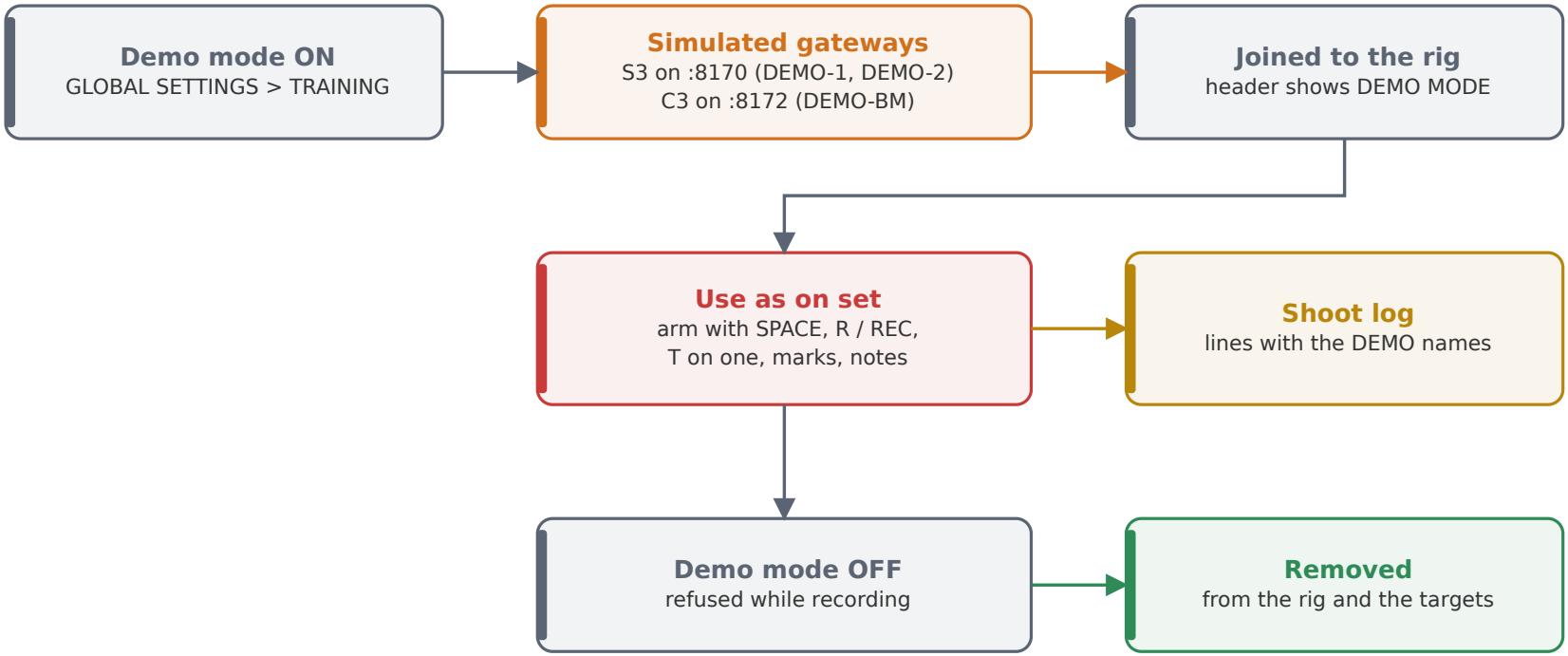
Services
Web page for others
USB controllers
on / off, kept

Warnings
camera / Atom battery
card space
disk space
bridge memory > 500 MB
C3 memory low
TC drift

- Warnings are raised once per condition and cleared when it recovers; they appear on the page and the console.

15 Demo mode

Practice the whole flow with no equipment: simulated gateways answer like the real ones.



- The simulators run inside the bridge on 127.0.0.1 only; nothing goes on the camera Wi-Fi.
- The DEMO gateway line in GATEWAYS is not a camera: the DEMO-1 / DEMO-2 / DEMO-BM rows are in the device list.

16 Code check: what was found and fixed

Static analysis (pyflakes, vulture, shellcheck), unused routes and JS, docs vs code, and a read of the risky paths.

FIXED	**Interfaces renamed by sed over bridge.py** Installer rewrote every matching word (it had damaged a comment). Now the names come from ap.conf.
FIXED	**SFTP send** No feedback; archives re-sent each time. Now progress bar, result, archives already on the server skipped.
FIXED	**Stale reads after a command** Page, console, GoPro, mobile, Canon: an old reply no longer undoes a new command.
FIXED	**Group REC with disarmed devices** Next press stops everything recording, even devices disarmed mid-take.
FIXED	**Per-device REC** Never stops other devices; second GoPro right after the first goes the right way.
FIXED	**Update looked stuck** Version is recorded; 'pulled but not installed' is detected and offered.
FIXED	**Console after update** Reopens by itself even with a page open.
FIXED	**Small leftovers** Unused import/constant, F6 notes as raw numbers, stale TODOs and a wrong Bluetooth comment.
OPEN	**Diagnostic tools in the repo root** ltc_gen.py, analyze_f6_take.py (old F6 Ext Auto Rec), f6_s3.py + s3_frc8/ (F6 via S3), f6_frc.py, probes. Kept on purpose; could move to tools/.
OPEN	**Bluetooth adapter chosen by number** hci0/hci1 change with hardware (ThinkPad). Choosing the dongle by its address would be safer.
OPEN	**Old 'Pi' wording in code comments** History only; no visible text says Pi. Route name /api/time/ltc/set_system_clock is internal.
TESTS	**111 automated tests** All pass locally and on GitHub on every commit.